



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SREE MALAR THANGAM AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

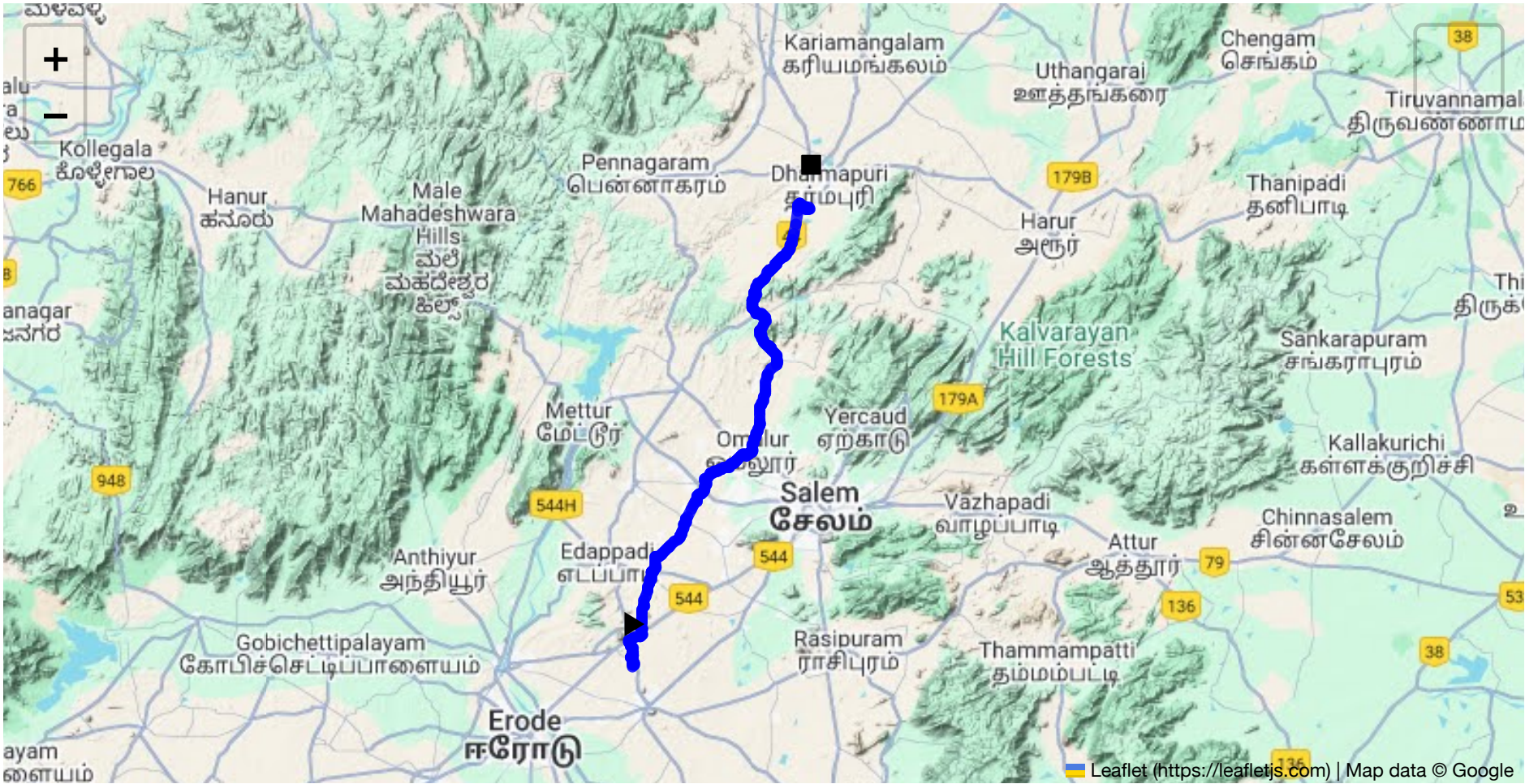


Route Summary:
Total Distance: 91.18 km
Estimated Duration: 2.1 hours

Adjusted Duration (Heavy Vehicle): 2.6 hours

Start: (11.4381, 77.8734)

End: (12.086568, 78.129255)



Welcome to the Journey Risk Management Study

- 1. Overview of the Route Map:** The route from Sangagiri to Jettihalli via Erumaipatti and Salem spans approximately 91.18 kilometers. It starts at Sangagiri, heads east towards Erumaipatti, deviates northward passing through Salem, and finally terminates at Jettihalli. The journey is on a mix of district roads and national highways. Key segments include NH44 from Salem, offering the best infrastructure.
- 2. Typical Weather Conditions and Potential Weather-Related Hazards:** Tamil Nadu generally has a tropical climate. The region experiences the northeast monsoon from October to December, potentially causing heavy rainfall and minor flooding, particularly in low-lying or poorly drained areas. During summer (March to May), expect high temperatures and occasional heat waves, while winter (December to February) is milder and more favorable for travel.
- 3. Analysis of Traffic Patterns:** Salem often experiences heavy traffic, especially during peak hours (8-10 AM and 5-8 PM). Congestion is frequent around urban centers and intersections, particularly near Salem due to city traffic. Weekends and festival days might see increased traffic going through Salem as it serves as a transit hub.
- 4. Assessment of Road Quality and Infrastructure:** Roads on this route vary in quality. National highways, particularly around Salem, are well-maintained. However, rural district roads may have uneven surfaces, with occasional potholes and narrow passages. These require cautious navigation, especially for heavy vehicles.
- 5. Suggestions for Alternative Routes for Emergencies:** In emergencies, consider bypassing Salem using the nearby SH188 to the west or the NH44 further towards the east, depending on the specific location and nature of the situation. Local roads around Erumaipatti to Rasipuram also offer detours but should only be used if necessary due to lesser road quality.
- 6. Summary of Local Regulations Affecting Hazardous Material Transport:** Tamil Nadu has regulations aligned with national standards concerning the transport of hazardous materials, requiring permits and

adherence to safety standards. Ensure all documentation is current, and that vehicles have appropriate signages and safety measures in place.

7. Overview of Historical Incidents: The region around Salem and major highways has seen accidents involving heavy vehicles, often due to overspeeding and reckless driving on congested highways. Specific incidents involving hazardous materials are rare but stress caution especially when approaching Salem.

8. Environmental Considerations and Sensitive Areas: There are few environmentally sensitive areas along this route, with agricultural lands being most common. Care should be taken to prevent contamination or spills, especially in the event of an accident.

9. Analysis of Communication Coverage: The entire route generally has good communication coverage; however, sporadic dead zones may exist in rural areas particularly between Erumaipatti and Salem. It is advisable to maintain communication checks regularly.

10. Estimated Emergency Response Times: In urban areas like Salem, emergency response can be relatively quick, within 15-20 minutes. In rural sections, expect longer response times of up to 45 minutes to an hour due to distance and road conditions.

12. Overall Summary of Risk Assessment: While the route is generally safe and well-travelled, specific risks arise from traffic congestion, especially in Salem, and variable road conditions on rural stretches. Weather impacts and clear communication should be prioritized, especially during monsoon. Adequate planning and awareness of detours and response plans mitigate these risks for hazardous material transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	Medium	11.44900, 77.87406	30 KM/Hr
3	Turn	Medium	11.45000, 77.87213	30 KM/Hr
4	Turn	High	11.45028, 77.87202	15 KM/Hr
5	Turn	Medium	11.45309, 77.87325	30 KM/Hr
6	Turn	Medium	11.46400, 77.87323	30 KM/Hr
7	Turn	Medium	11.46765, 77.87037	30 KM/Hr
8	Turn	High	11.47403, 77.86841	15 KM/Hr
9	Turn	Medium	11.48363, 77.88780	30 KM/Hr
10	Turn	Medium	11.49182, 77.88762	30 KM/Hr
11	Turn	High	11.49451, 77.88489	15 KM/Hr
12	Turn	Medium	11.56092, 77.89927	30 KM/Hr
13	Turn	Medium	11.56191, 77.90024	30 KM/Hr
14	Turn	High	11.57035, 77.90224	15 KM/Hr
15	Turn	High	11.57040, 77.90222	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
16	Turn	Medium	11.58266, 77.90658	30 KM/Hr
17	Turn	Medium	11.63832, 77.94818	30 KM/Hr
18	Turn	Medium	11.64161, 77.95177	30 KM/Hr
19	Turn	Medium	11.64226, 77.95200	30 KM/Hr
20	Turn	Medium	11.64716, 77.95118	30 KM/Hr
21	Turn	High	11.67868, 77.96836	15 KM/Hr
22	Turn	Medium	11.70077, 77.97957	30 KM/Hr
23	Turn	High	11.70223, 77.97860	15 KM/Hr
24	Turn	Medium	11.71918, 78.01127	30 KM/Hr
25	Turn	Medium	11.72565, 78.01770	30 KM/Hr
26	Turn	Medium	11.73769, 78.03451	30 KM/Hr
27	Turn	Medium	11.73785, 78.03551	30 KM/Hr
28	Turn	High	11.73651, 78.03871	15 KM/Hr
29	Turn	Medium	11.74271, 78.04717	30 KM/Hr
30	Turn	Medium	11.87942, 78.08197	30 KM/Hr
31	Turn	Medium	11.95818, 78.04978	30 KM/Hr
32	Turn	Medium	11.95876, 78.04990	30 KM/Hr
33	Turn	Medium	11.96035, 78.04946	30 KM/Hr
34	Turn	Medium	11.96304, 78.05136	30 KM/Hr
35	Turn	Medium	11.96399, 78.05135	30 KM/Hr
36	Turn	Medium	11.96458, 78.05084	30 KM/Hr
37	Turn	Medium	11.96496, 78.04988	30 KM/Hr
38	Turn	High	11.96580, 78.04923	15 KM/Hr
39	Turn	Medium	11.96675, 78.04939	30 KM/Hr
40	Turn	Medium	11.96791, 78.05070	30 KM/Hr
41	Blind Spot	Blind Spot	12.09201, 78.11462	10 KM/Hr
42	Turn	High	12.08762, 78.12236	15 KM/Hr
43	Turn	High	12.08688, 78.12234	15 KM/Hr
44	Turn	High	12.08464, 78.12765	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
7	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
8	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
10	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
11	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
12	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
13	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
14	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
15	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
16	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
17	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
19	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
20	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
21	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
22	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
23	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
24	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
25	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
26	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
27	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
28	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
29	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
30	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
31	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
33	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
34	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
35	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
36	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
18	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
32	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
37	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45000, 77.87213



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



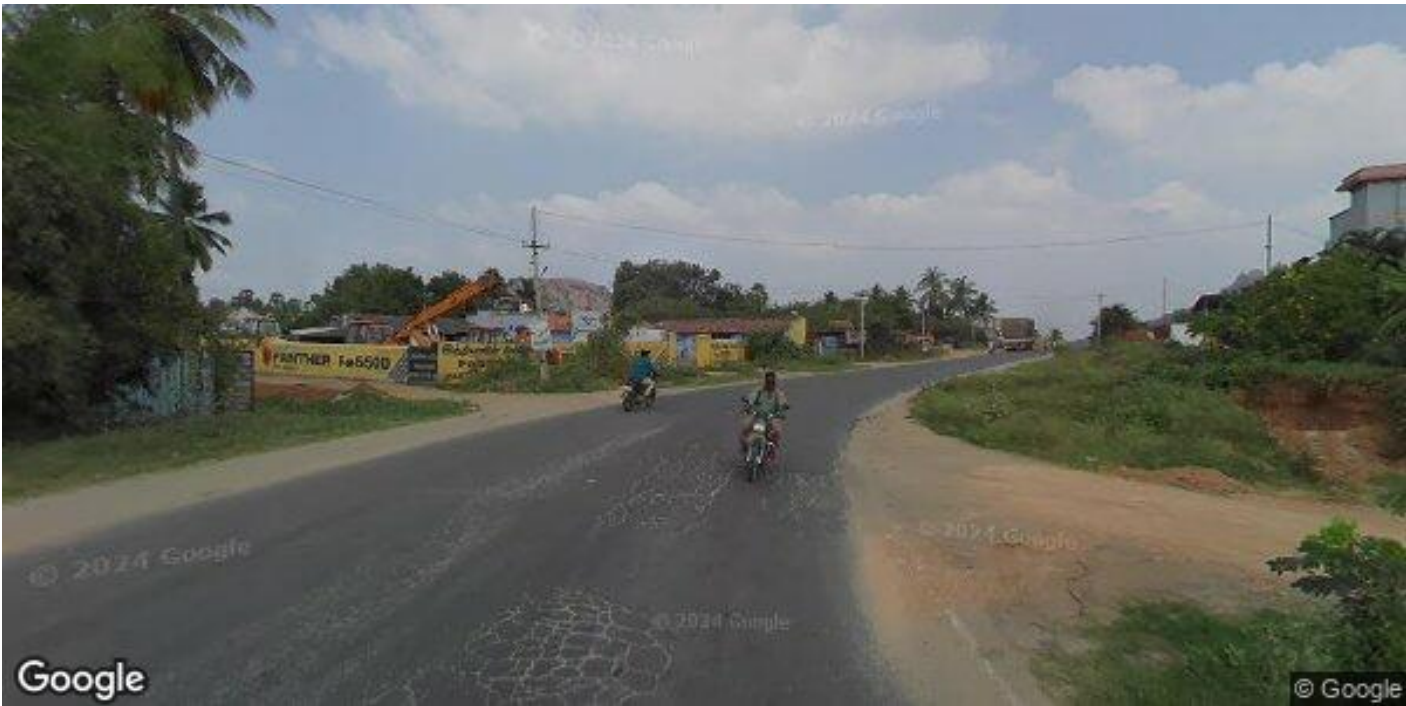
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48363, 77.88780



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49182, 77.88762



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49451, 77.88489



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56092, 77.89927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56191, 77.90024



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57035, 77.90224



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57040, 77.90222



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58266, 77.90658



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63832, 77.94818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64161, 77.95177



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64226, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64716, 77.95118



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67868, 77.96836



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70077, 77.97957



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70223, 77.97860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71918, 78.01127



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72565, 78.01770



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73769, 78.03451



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73785, 78.03551



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74271, 78.04717



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95876, 78.04990



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96035, 78.04946



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96458, 78.05084



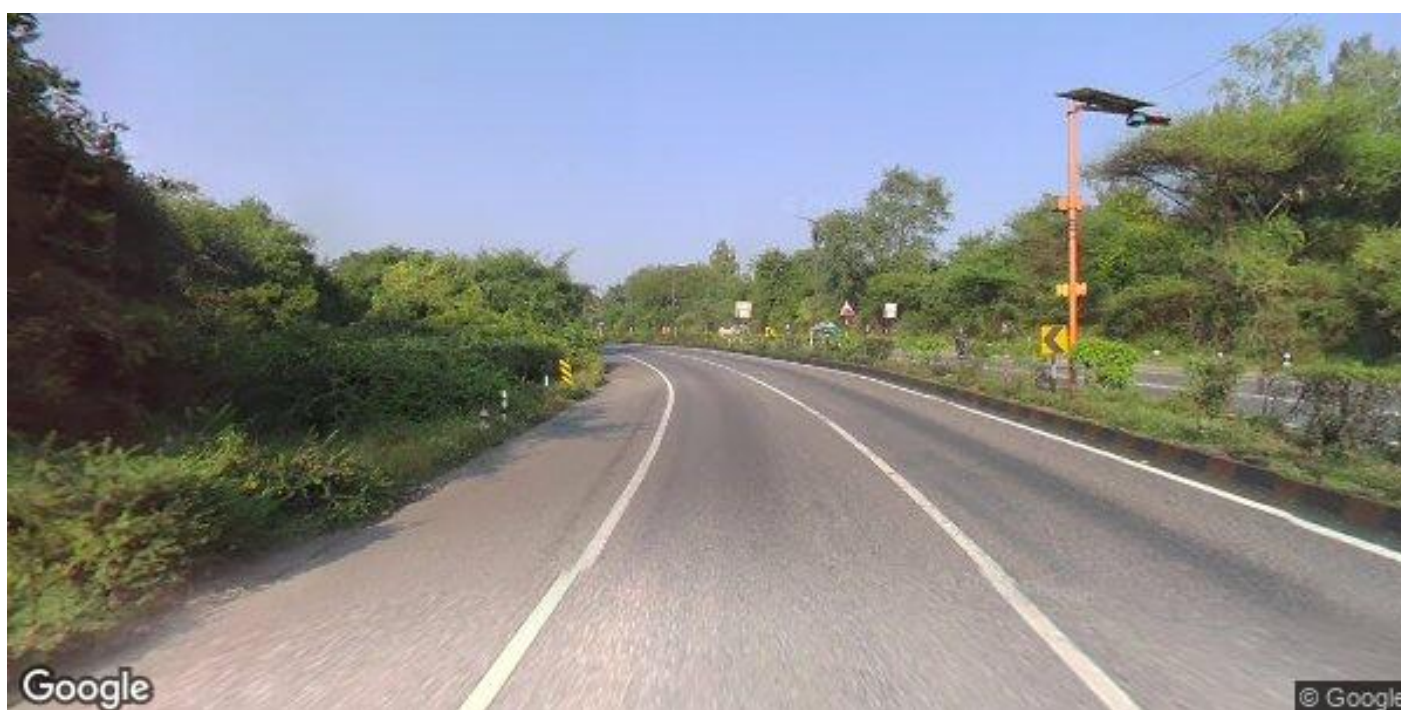
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96496, 78.04988



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.09201, 78.11462



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.08762, 78.12236



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.08688, 78.12234



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.08464, 78.12765

